Solar Empowerment

by Tracy Fox

Gary Gerber, CEO and founder of Sun Light and Power in Berkeley, California, has been working in the renewable energy field for 30 years. He successfully steered his solar design and installation company through the lean years when the country all but abandoned solar energy to the more recent second solar boom times. His company is thriving well enough to have been named one of the top 100 fastest-growing companies in the Bay Area by the *San Francisco Times* in 2004, 2005, and 2006, and he expects it to be on the list again in 2007. With over 40 employees, Sun Light and Power (SLP) is at once a pioneer and an expert in the field of solar design and installation. "I am a general contractor involved in sustainability, and I've been traveling down the road of green construction for over ten years, with a keen interest in the bigger picture of energy," says Gerber.

SLP has amassed an impressive portfolio of over 600 installations throughout the Bay Area; it is an industry leader in solar-thermal and solar-electric design and installation. SLP employees are also passionate about the company's overall environmental commitment to promote renewable energy sources. "Many of our installers volunteer their time with Grid Alternatives doing one or two systems a month. These are weekend installations of free solar systems throughout the Bay Area, installing systems for Habitat for Humanity as well as other low-income housing providers," says Gerber. The company's motto: "Changing the world one roof at a time."

THE FUTURE OF ENERGY

Gerber remembers the specific moment when he realized how important it is to preserve the world's vital resources. It was the early '70s, and he was at the University of California, Berkeley, where he was studying energy and heat transfer. "We had a guest speaker come in and talk about the different forms of energy being used throughout the world, and he said there is about enough coal for 200 years and enough oil for 50 or maybe 100 years. When you think about these things in geological terms and even human terms, it is a blink of an eye," says Gerber. "Solar and wind power stood out as the only energy technologies that had any real future."


POWERLIGHT CORP

Sun Light and Power's company motto is "Changing the world one roof at a time."

Today SLP offers the latest technological advancements in solar electricity, solar hot water, and radiant heating, with half of their business coming from residential and half from commercial projects. "In many cases, those customers find us through word of mouth," explains Gerber. "We make a real effort to develop relationships with repeat customers—building professionals such as architects, builders, roofers, and housing developers. West Coast Green [a residential green building conference] was also a huge success for us, and we do well at other targeted home shows too."

While skyrocketing energy prices have encouraged people to turn to solar energy, Gerber finds that people are looking at the bigger picture, not just at their pocketbooks. "I do find that global warming is on the minds of almost all of my clientele. They recognize there is a problem, and they want to be part of the solution," says Gerber. He also sees a stronger interest in solar energy in California, especially in Northern California, than in other parts of the country, and statistics confirm this impression. Since early 2005, California has seen a growing influx of venture capital money; in the first six months of 2006, $1.6 billion was invested in clean-tech companies, compared to $1.4 billion in all of 2005.

TOTAL ENERGY SOLUTIONS

The current solar renaissance is due in part to the passage in 2006 of the largest solar incentive program to date—the California Solar Initiative (CSI) (see "The California Solar Initiative," p. 6). This ambitious ten-year, $2.9 billion program seeks to install 3,000 MW of rooftop solar systems by 2017. However, the new rebate program has more restrictive rules than the old one, including a requirement for energy audits before qualification for a rebate, Gerber points out, and rebates are lower than they were before, making it harder to sell solar systems. "What is going on now is that the CSI is requiring energy audits from everyone who wants to apply for a rebate, and I believe this will move the industry in the right direction in energy efficiency," says Gerber, whose initial approach includes working with the network of raters from Build It Green to measure home energy performance.

While most solar companies don't offer any additional services beyond solar installations, energy efficiency contracting is already an emerging part of the SLP business model—and Gerber expects this part of his business to grow. He plans to offer his clients a package of what he calls Total Energy Solutions, identifying the most egregious inefficiencies in a home and recommending the solutions that have the best payback. "The best approach is to reduce your demand first, before you start spending a lot of money on a solar system to satisfy a demand that is too high to begin with," he says.

In order for a PV system on a new home to qualify for the CSI rebate, the new home must exceed California's Title 24 energy efficiency standards by 15%. "Solar hot water is one of the best and easiest ways to take your home above and beyond Title 24," says Gerber. While enthusiasm for the CSI is strong, currently there is still uncertainty in how to meet some of the CSI's new rebate requirements, making the new program somewhat of a trial from a sales and promotion perspective. "It [the solar industry] is going to be a robust and growing industry, but it is a much more challenging industry today than it was even a month ago to sell systems in," says Gerber.

THREE STRIKES

To prepare for meeting the CSI's new, more complicated requirements, SLP hired a dedicated rebate assistant for its full-time contract manager, Sarah Diaz. "Keeping on top of rebate issues for your company, if you are not big enough to have a dedicated person in that role already, is going to be difficult and make it harder to avoid getting strikes," Diaz says. Yes, strikes. Any sign of fraud or incompetence in installing systems and applying for rebates counts as a strike against the installing company, and three strikes means that the company is locked out of rebate applications, says Gerber. Even though the majority of companies that design and install solar systems have done a high-quality job, there are always a few, as in any industry, that haven't. However, these new requirements can affect even the good players, according to Gerber, who sees routine inspections not only as the best way to identify people who are defrauding the system, but also as the way to identify errors made by legitimate businesses. "There is a big difference between intentional fraud and the occasional goof, and you don't want to penalize the business that has every intention of getting it right but goofs once in awhile," says Gerber.

A strike expires after a year for companies that install 200 or more systems per year. SLP, a company with an excellent reputation for quality, installed nearly 200 systems last year. One problem for high-volume businesses, Gerber worries, is that any company that does enough business might eventually pick up a few strikes just by the law of averages. A worker could make a mistake on the job site or fill in the wrong number on a form, and then the company would be stuck with a strike. And the bottom line is that if a company is excluded from the CSI for one year, that company is effectively out of the solar business in California. It remains to be seen how to make the three-strike requirement practical for all solar installers. The California Public Utilities Commission has created a Solar Forum to hold regular meetings with all interested parties to hammer out these issues over the life of the program.

MORE GROWING PAINS

The CSI and the new rules that go with it aren't the only thing that is new—even the old familiar design tools are taking on new forms. In the past, SLP solar designers have used PVWatts, an online performance calculator for grid-connected PV systems, says Gerber. Going forward, anyone who wants a rebate on solar energy in California will have to use the CSI Expected Performance Based Buydown Calculator. It is necessary to use this standardized design tool because the rebate has to be applied consistently to all systems now, explains Gerber.

Looking ahead, Gerber likes the fact that the ten-year guaranteed life of the CSI program will encourage more solar manufacturers to jump into the game. This is good news for California's solar industry and good news for the state's economy, although solar companies will still have to learn to navigate through uncharted territory. "I think what is going to happen with the solar industry is, we are going to need to get more knowledgeable and effective with our energy efficiency," Gerber says. That's a good thing. With Gerber at the helm, Sun Light and Power will no doubt continue to be a leader in solar-energy systems.

Tracy Fox writes about energy efficiency, sustainable design, and building practices. Her firm, Foxline Design, is based in Santa Rosa, California.

For more information:

Sun Light and Power
1035 Folger Ave.
Berkeley, CA 94710
(510)845-2997
www.sunlightandpower.com

For more on Build It Green, go to www.builditgreen.org.